5 **(a)** When monochromatic radiation is incident on a clean cadmium surface, electrons with a range of kinetic energies up to a maximum of 3.65×10^{-20} J are released. The work function of cadmium is 4.07 eV.

(i) Explain what is meant by *work function*.

....

.... [1]

(ii) Explain why the emitted electrons have a range of kinetic energies up to a maximum.

....

....

....

.... [2]

(iii) Calculate the wavelength of the radiation.

wavelength = ... m [2]

(b) In the wave model of electromagnetic radiation, electrons near the surface of a metal absorb energy from the wave.

(i) The radiation incident on the cadmium surface in **(a)** delivers energy at a rate of 3.0×10^{-22} J s^{-1}.

Calculate the minimum time it would take an electron to absorb sufficient amount of energy to be emitted from the metal surface.

time = ... s [1]

(ii) In the actual experiment conducted, photoelectrons are observed to be emitted without any time lag.

Explain how this observation provides evidence for the particulate nature of electromagnetic radiation.

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…………………………………………………………………………………………… [2]

6 Tritium is an isotope of hydrogen and is represented by the symbol ^{3}H